

Bayesian Inference

From Belief Updating to Posterior Predictives
A Graduate Introduction

Statistical Methods Seminar

Department of Statistics

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Outline

Probability as Belief

Prior, Likelihood, and Posterior

Conjugate Priors

Point Estimates & Credible Intervals

Hierarchical Models

Markov Chain Monte Carlo

Hamiltonian Monte Carlo

Posterior Predictive Checking

Model Comparison

The Bayesian Workflow

Summary

Two Interpretations of Probability

Frequentist

- Probability = long-run frequency
- Parameters are *fixed* unknowns
- Data are random
- No probability statements about θ
- Confidence intervals, p -values

Data $\xrightarrow{\text{estimator}}$ $\hat{\theta}$
Point estimate only

vs.

Bayesian

- Probability = *degree of belief*
- Parameters have probability distributions
- Prior $\rightarrow$ Posterior via data
- Full uncertainty quantification
- Credible intervals, Bayes factors

Prior + Data $\xrightarrow{\text{Bayes}}$ $p(\theta | \mathcal{D})$
Full distribution

Bayes' Theorem — The Engine

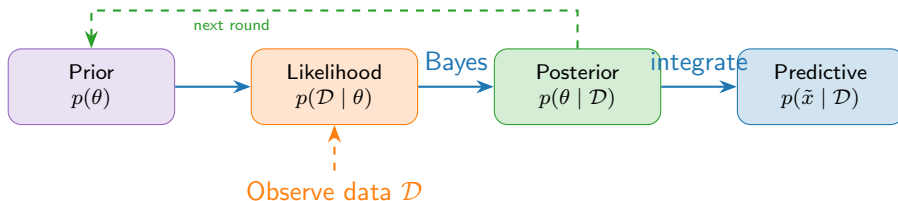
Bayes' Theorem

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{marginal likelihood}}}$$

- θ — parameter(s) of interest
- $\mathcal{D} = (x_1, \dots, x_n)$ — observed data
- $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta) p(\theta) d\theta$ — normalising constant (often *intractable*)
- **Key insight:** $p(\mathcal{D})$ does not depend on θ , so

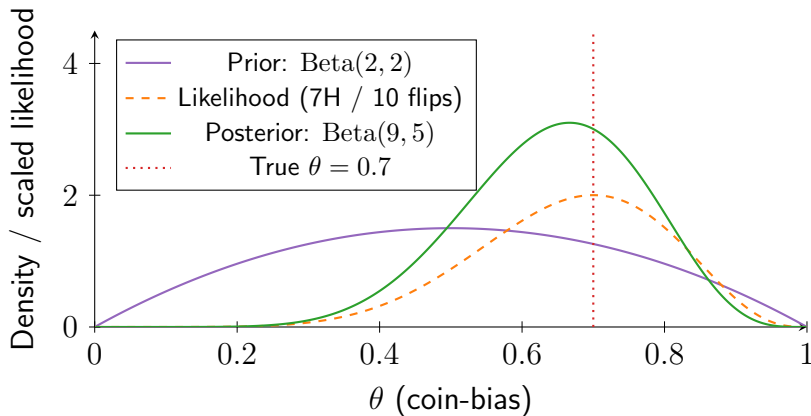
$$\boxed{p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta)} \Leftrightarrow \text{posterior} \propto \text{likelihood} \times \text{prior}$$

Sequential Belief Updating



- Bayesian learning is **sequential**: today's posterior is tomorrow's prior
- For i.i.d. data, order of observations does not matter

Coin-Flip Example: Prior, Likelihood, Posterior



Prior Beta(2, 2) + 7 heads in 10 flips $\Rightarrow$ Posterior Beta(9, 5). The posterior is pulled toward the data (MLE = 0.7).

Conjugate Priors — Closed-Form Posteriors

Definition (Conjugate Family)

A prior $p(\theta)$ is **conjugate** to likelihood $p(\mathcal{D} \mid \theta)$ if the posterior $p(\theta \mid \mathcal{D})$ belongs to the *same parametric family* as the prior.

Likelihood	Parameter	Conjugate Prior	Posterior
Binomial(n, θ)	θ	Beta(α, β)	Beta($\alpha + k, \beta + n - k$)
Poisson(λ)	λ	Gamma(a, b)	Gamma($a + \sum x_i, b + n$)
$\mathcal{N}(\mu, \sigma^2)$	μ (σ^2 known)	$\mathcal{N}(\mu_0, \sigma_0^2)$	$\mathcal{N}(\mu_n, \sigma_n^2)$
Multinomial	$\boldsymbol{p}$	Dir($\boldsymbol{\alpha}$)	Dir($\boldsymbol{\alpha} + \boldsymbol{n}$)

Normal update (precision form)

Beta–Binomial: A Worked Derivation

Setup: k heads in n flips; model $\theta \sim \text{Beta}(\alpha, \beta)$.

Derivation:

$$\begin{aligned} p(\theta \mid k) &\propto p(k \mid \theta) p(\theta) \\ &= \binom{n}{k} \theta^k (1 - \theta)^{n-k} \cdot \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)} \\ &\propto \theta^{\alpha+k-1} (1 - \theta)^{\beta+n-k-1} \end{aligned}$$

$$p(\theta \mid k) = \text{Beta}(\alpha + k, \beta + n - k)$$

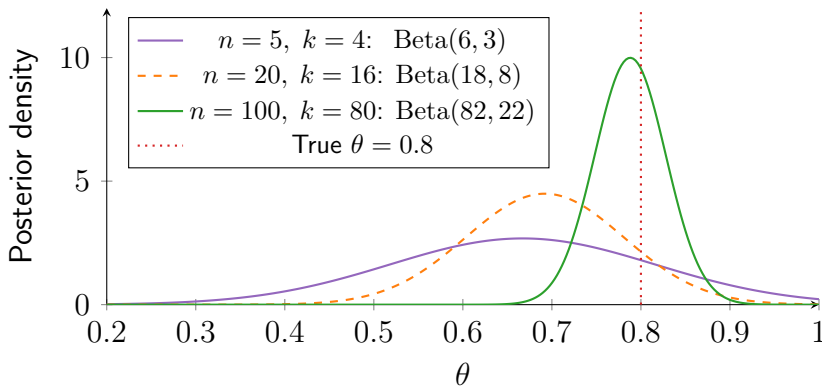
Posterior mean:

$$\mathbb{E}[\theta \mid k] = \frac{\alpha + k}{\alpha + \beta + n} = \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} \cdot \lambda + \underbrace{\frac{k}{n}}_{\text{MLE}} \cdot (1 - \lambda)$$

Interpretation:

- α, β act as *pseudo-counts*
- As $n \rightarrow \infty$: posterior $\rightarrow$ MLE
- $\alpha + \beta =$ effective prior sample

Prior Sensitivity: Data Overwhelms the Prior



Starting from the same prior Beta(2, 2): as n grows the **likelihood dominates**, and posteriors converge regardless of prior choice.

Summaries of the Posterior

Point Estimates

- **MAP** (Maximum a Posteriori):

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \mathcal{D})$$

- **Posterior Mean:**

$$\hat{\theta}_{\text{PM}} = \mathbb{E}[\theta \mid \mathcal{D}]$$

- **Posterior Median:** minimises expected absolute loss

Credible Intervals

- A $(1 - \alpha)$ **credible interval** C :

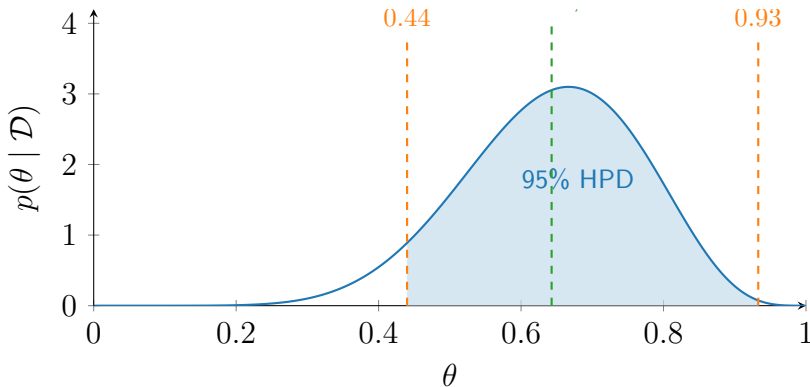
$$P(\theta \in C \mid \mathcal{D}) = 1 - \alpha$$

- **HPD** (Highest Posterior Density): shortest such interval
- *Direct* probability statement about θ (unlike frequentist CI!)

Frequentist vs. Bayesian interval

A 95% confidence interval does *not* mean “95% probability θ lies here.” A 95%

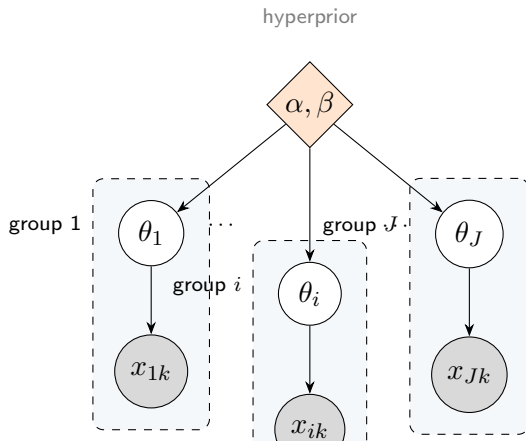
HPD Interval — TikZ / pgfplots Illustration



Posterior $\text{Beta}(9, 5)$; 95% HPD $\approx [0.44, 0.93]$. Posterior mean $= 9/14 \approx 0.643$. Shaded region contains exactly 95% of posterior probability.

Hierarchical (Multi-Level) Models

When data have **grouped structure**, group-level parameters themselves have parameters (hyperparameters).



Shrinkage: Partial Pooling

- **No pooling:** estimate each group independently

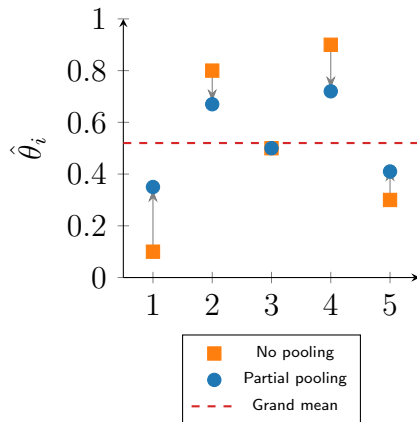
$$\hat{\theta}_i = \frac{k_i}{n_i}$$

- **Complete pooling:** ignore group structure

$$\hat{\theta} = \frac{\sum k_i}{\sum n_i}$$

- **Partial pooling** (hierarchical):

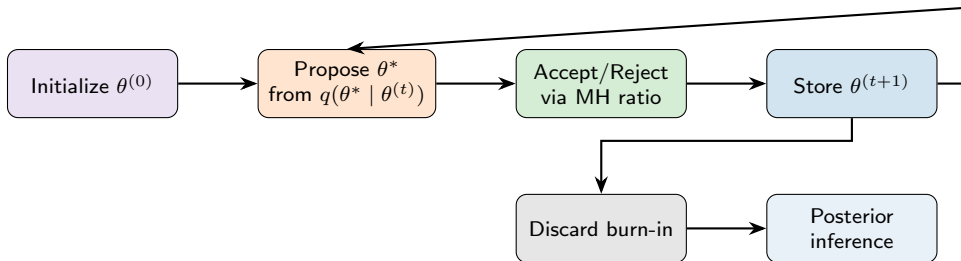
$$\hat{\theta}_i \approx \lambda_i \frac{k_i}{n_i} + (1 - \lambda_i) \bar{\theta}$$



When Conjugacy Fails: MCMC

For complex models $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta) p(\theta) d\theta$ is **intractable**.

Strategy: Draw correlated samples $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta \mid \mathcal{D})$ without evaluating $p(\mathcal{D})$.



Metropolis–Hastings Algorithm

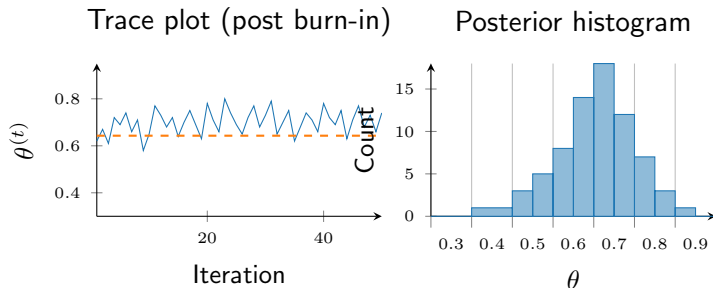
MH Algorithm

1. Start at $\theta^{(0)}$. For $t = 1, 2, \dots, S$:
2. **Propose**: draw $\theta^* \sim q(\theta^* \mid \theta^{(t-1)})$
3. **Acceptance ratio**:

$$r = \frac{p(\mathcal{D} \mid \theta^*) p(\theta^*)}{p(\mathcal{D} \mid \theta^{(t-1)}) p(\theta^{(t-1)})} \cdot \frac{q(\theta^{(t-1)} \mid \theta^*)}{q(\theta^* \mid \theta^{(t-1)})}$$

4. **Accept**: set $\theta^{(t)} = \theta^*$ with probability $\min(1, r)$; else $\theta^{(t)} = \theta^{(t-1)}$
- Under mild conditions, the chain converges in distribution to $p(\theta \mid \mathcal{D})$
 - Symmetric proposal q (e.g. $\mathcal{N}(0, \sigma_q^2)$) simplifies r to ratio of posterior densities
 - **Tuning**: proposal variance reverses acceptance rate (target 23–44%)

MCMC: Trace Plot and Posterior Histogram



Left: a well-mixed trace looks like “hairy caterpillar” white noise — no drift. Orange dashed line = posterior mean $9/14 \approx 0.643$.

Right: histogram of samples approximates $p(\theta \mid \mathcal{D})$. MCMC diagnostics: $\hat{R} < 1.01$, $\text{ESS} > 400$.

Beyond Random Walk: Hamiltonian Monte Carlo (HMC)

Problem with MH: random-walk proposals are inefficient in high dimensions (slow mixing, high autocorrelation).

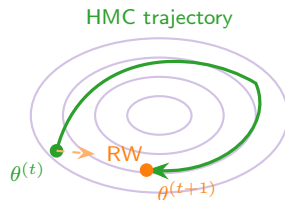
HMC idea: augment θ with auxiliary momentum ρ and simulate Hamiltonian dynamics:

$$H(\theta, \rho) = \underbrace{-\log p(\theta \mid \mathcal{D})}_{\text{potential energy } U(\theta)} + \underbrace{\frac{1}{2}\rho^\top M^{-1}\rho}_{\text{kinetic energy}}$$

Leapfrog integrator (one step):

$$\begin{aligned}\rho_{t+\epsilon/2} &= \rho_t - \frac{\epsilon}{2} \nabla_{\theta} U(\theta_t) \\ \theta_{t+\epsilon} &= \theta_t + \epsilon M^{-1} \rho_{t+\epsilon/2} \\ \rho_{t+\epsilon} &= \rho_{t+\epsilon/2} - \frac{\epsilon}{2} \nabla_{\theta} U(\theta_{t+\epsilon})\end{aligned}$$

NUTS: No-U-Turn Sampler
auto-tunes the trajectory length



Posterior Predictive Distribution

Given new input $\tilde{x}$, what is our probabilistic forecast?

$$p(\tilde{x} \mid \mathcal{D}) = \int p(\tilde{x} \mid \theta) p(\theta \mid \mathcal{D}) d\theta$$

- Averages over posterior uncertainty in θ (more honest than plugging in $\hat{\theta}$)
- Analytical for conjugate models (e.g. Beta–Binomial predictive)
- Approximated via MCMC samples: $p(\tilde{x} \mid \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S p(\tilde{x} \mid \theta^{(s)})$

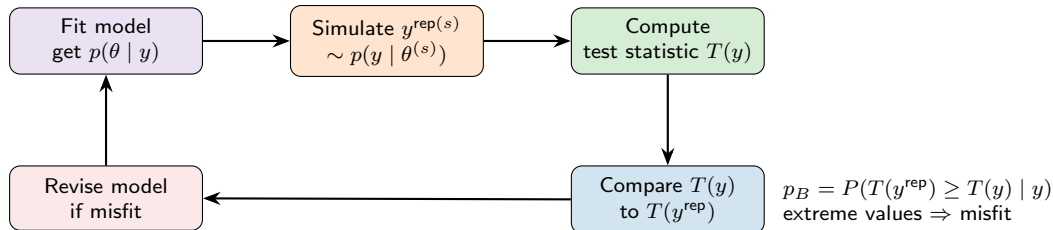
Beta–Binomial Predictive (m future trials)

With $\theta \mid \mathcal{D} \sim \text{Beta}(a, b)$:

$$p(\tilde{k} \mid \mathcal{D}) = \binom{m}{\tilde{k}} \frac{B(a + \tilde{k}, b + m - \tilde{k})}{B(a, b)}, \quad \tilde{k} = 0, \dots, m$$

Posterior Predictive Check (PPC)

Goal: Detect model misspecification by comparing replicated data y^{rep} to observed y .



Bayesian Model Comparison

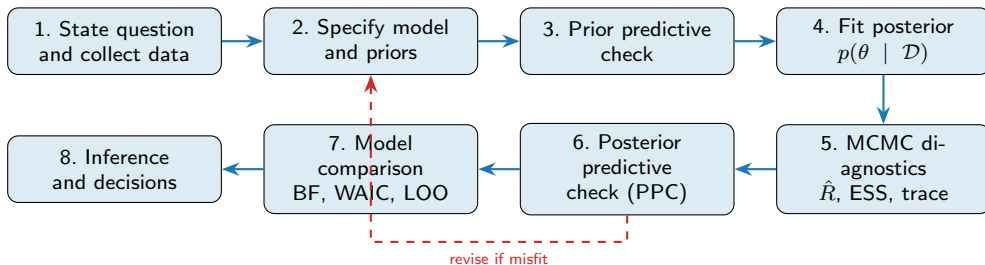
Given models $\mathcal{M}_1, \mathcal{M}_2$, compare via the **Bayes factor**:

$$\text{BF}_{12} = \frac{p(\mathcal{D} \mid \mathcal{M}_1)}{p(\mathcal{D} \mid \mathcal{M}_2)} = \frac{\int p(\mathcal{D} \mid \theta_1, \mathcal{M}_1) p(\theta_1 \mid \mathcal{M}_1) d\theta_1}{\int p(\mathcal{D} \mid \theta_2, \mathcal{M}_2) p(\theta_2 \mid \mathcal{M}_2) d\theta_2}$$

BF_{12}	Evidence for $\mathcal{M}_1$
1–3	Anecdotal
3–10	Moderate
10–30	Strong
> 100	Decisive

- BF naturally penalises complexity (**Occam's razor** / marginal likelihood)
- Posterior model probability: $P(\mathcal{M}_k \mid \mathcal{D}) \propto p(\mathcal{D} \mid \mathcal{M}_k) P(\mathcal{M}_k)$
- Practical alternatives: **WAIC**, **LOO-CV** (computationally cheaper, no closed-form marginal needed)

Bayesian Workflow — End to End



Bayesian modelling is iterative: predictive checks and diagnostics can send us back to revise the model, priors, or data-generating assumptions.

Key Takeaways

1. **Posterior** $\propto$ **likelihood** $\times$ **prior**:

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta)$$

2. **Conjugate priors** yield closed-form posteriors; useful for intuition and simple analytics.
3. **Credible intervals** have direct probability interpretations; hierarchical models share information via partial pooling.
4. **MCMC** (MH, HMC/NUTS) enables inference for arbitrary models; always run diagnostics ($\hat{R}$, ESS).
5. **PPC** and **LOO-CV** check model adequacy; iterate the workflow until checks pass.

Recommended Software

Stan (NUTS, gradient-based) · **PyMC** (Python, flexible) · **NumPyro**
(CRUI friendly, JAX) · **Turing.jl** (Julia)

Appendix: Key Distribution Reference

Distribution	Parameters	Mean	Variance
Beta(α, β)	$\alpha, \beta > 0$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$
Gamma(a, b)	$a, b > 0$	a/b	a/b^2
$\mathcal{N}(\mu, \sigma^2)$	$\mu \in \mathbb{R}, \sigma^2 > 0$	μ	σ^2
Inv-Gamma(α, β)	$\alpha > 2$	$\frac{\beta}{\alpha - 1}$	$\frac{\beta^2}{(\alpha - 1)^2(\alpha - 2)}$

Appendix: MCMC Diagnostics

Gelman–Rubin $\hat{R}$

- Compare within-chain vs. between-chain variance
- $\hat{R} < 1.01$ indicates convergence
- Requires ≥ 2 chains with dispersed starts

Effective Sample Size (ESS)

$$\text{ESS} = \frac{S}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

- $\rho_k = \text{lag-}k \text{ autocorrelation}$

Visual checks

- Trace plots: “hairy caterpillar”
- Rank plots: uniform across chains
- Autocorrelation function (ACF): fast decay

Divergences (HMC)

- Divergent transitions indicate geometry problems
- Fix: increase `adapt_delta`, reparameterise (non-centred)

Appendix: Recommended Reading

- **Gelman et al.** (2013). *Bayesian Data Analysis*, 3rd ed. (BDA3).
- **McElreath** (2020). *Statistical Rethinking*, 2nd ed. (accessible, causal reasoning).
- **Bernardo & Smith** (2000). *Bayesian Theory* (theoretical foundations).
- **Robert** (2007). *The Bayesian Choice* (decision-theoretic perspective).
- **Stan Development Team** (2023). *Stan Reference Manual*. mc-stan.org